

$\text{Li}_x\text{CoO}_2$  ( $0 < x \leq 1$ ): A NEW CATHODE MATERIAL FOR BATTERIES OF HIGH ENERGY DENSITY

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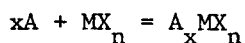
ABSTRACT

A new system  $\text{Li}_x\text{CoO}_2$  ( $0 < x \leq 1$ ) has been prepared by electrochemical extraction of lithium from the parent  $\text{LiCoO}_2$  ordered rock-salt structure. Measured open-circuit voltages with respect to lithium metal are approximately twice those of  $\text{Li}_x\text{TiS}_2$  ( $0 \leq x \leq 1$ ), and a theoretical energy density of  $1.11 \text{ kWh kg}^{-1}$  is calculated for  $\text{Li}_x\text{CoO}_2/\text{Li}$ . Preliminary voltage-composition curves show low overvoltages and good reversibility for current densities up to  $4 \text{ mA cm}^{-2}$  over a large range of  $x$ .

Introduction

The feasibility of insertion-compound electrodes for nonaqueous secondary batteries of high specific energy and power was first demonstrated with the  $\text{Li}_x\text{TiS}_2/\text{Li}$  cell in 1976 (1). Lithium was reversibly inserted into and extracted from a  $\text{TiS}_2$  positive electrode  $\text{Li}_x\text{TiS}_2$  over the compositional range  $0 \leq x \leq 1$ ; over most of this range a cell voltage in excess of 2 V was maintained at current densities of  $1\text{--}10 \text{ mA cm}^{-2}$ . However, difficulties with practical nonaqueous electrolytes and reversible deposition at the lithium negative electrode have restricted commercial  $\text{Li}_x\text{TiS}_2/\text{Li}$  cells to "button-size" units. Any change in the negative electrode from pure lithium to alleviate these problems reduces the voltage of the cell. Therefore, there is a motivation to obtain a positive electrode giving a larger cell voltage relative to lithium than does  $\text{Li}_x\text{TiS}_2$  while retaining a comparable range of solid solution.

A larger negative free energy change for a reaction



is expected when A is small and electropositive,  $\text{MX}_n$  contains a metal atom M in a high oxidation state, and X is small and electronegative. Therefore, for a higher cell voltage, lithium is preferable to sodium, and oxides are preferable to sulphides. Moreover, a higher oxidation state of M is more likely to be thermodynamically stable as an oxide than as a sulphide. The problem is to prepare electronically conducting oxides containing M in a high

oxidation state with structures suitable for fast-ion insertion.

Since 1976, many insertion compounds have been explored; they can all be synthesized by electrochemical lithium insertion, the method originally used for  $\text{Li}_x\text{TiS}_2$ . The materials studied include lithium-intercalated sulphides such as  $\text{MoS}_2$  and  $\text{NiPS}_3$ , oxides such as  $\text{V}_2\text{O}_5$  and  $\text{MoO}_3$ , and the oxyhalide  $\text{FeOCl}$  (2). In addition to these layer compounds, the oxide  $\text{V}_6\text{O}_{13}$  has also been investigated (3). The purpose of the investigation reported here was twofold: (1) to demonstrate a low-temperature synthetic route to oxides containing high valence states of heavier transition-metal ions of a long series and (2) to determine whether  $\text{Li}^+$ -ion mobilities in layered, metallic oxides can be high enough to sustain large voltages reversibly at current densities of the order of  $1 \text{ mA cm}^{-2}$ .

The synthetic route chosen was to prepare the discharged-electrode material, a ternary lithium oxide, with high-temperature techniques and to obtain the charged material by low-temperature electrochemical extraction of the lithium. Previous syntheses have used high-temperature techniques to obtain the oxidized compound of a fully charged electrode and low-temperature electrochemical insertion of lithium during cell discharge. However, the chosen synthetic route requires use of a starting high-temperature material that allows low-temperature extraction of lithium. The resultant phase need not necessarily be thermodynamically stable.

Examination of the known  $\text{Li}^+$ -ion solid electrolytes led us to conclude that  $\text{Li}^+$  ions may be mobile in close-packed oxygen arrays. Given that hypothesis, it was then natural to consider the transition-metal lithium oxides  $\text{LiMO}_2$  crystallizing in the layered structure illustrated in Fig. 1. The  $\text{Li}^+$  and  $\text{M}^{3+}$  ions occupy alternate (111) layers of the rocksalt structure; removal of the  $\text{Li}^+$  ions would generate an  $\text{MO}_2$  phase having the layered  $\text{CdCl}_2$  structure. We undertook to investigate how much lithium could be removed reversibly.

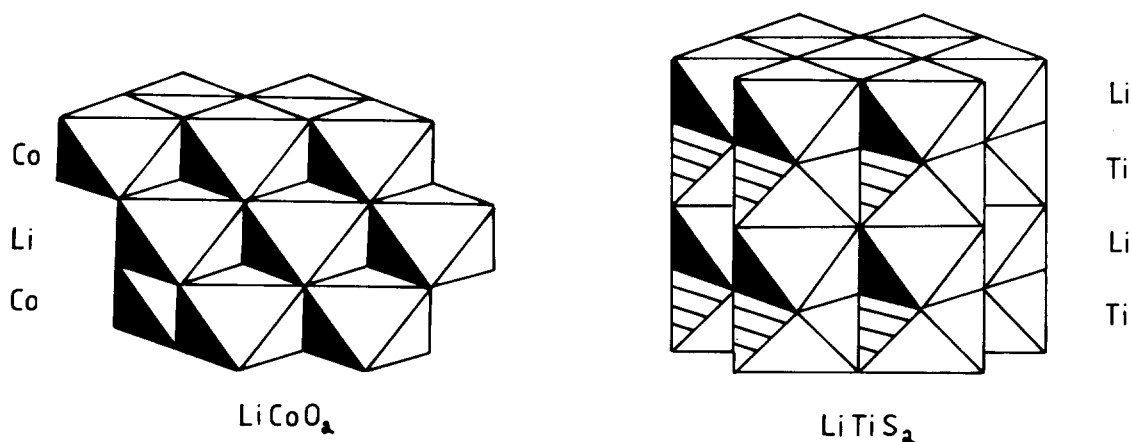


FIG. 1  
Idealised structures of  $\text{LiCoO}_2$  and  $\text{LiTiS}_2$

The essential difference between the  $\text{CdCl}_2$  structure and the  $\text{CdI}_2$  structure of  $\text{TiS}_2$  is in the close-packed arrangement of the anions. In  $\text{CdCl}_2$ , the anions form a close-packed pseudocubic array; in  $\text{CdI}_2$  they form a close-packed hexagonal array. The layered rocksalt structure is adopted for several transition-metal lithium oxides  $\text{LiMO}_2$ , including  $\text{M} = \text{V}$  (4),  $\text{Cr}$  (5),  $\text{Co}$  (6), and  $\text{Ni}$  (7). We have also obtained it for  $\text{M} = \text{Fe}$  by low-temperature ion exchange of  $\text{NaFeO}_2$ . We report here our results for  $\text{LiCoO}_2$ , which appears to be the most promising of these systems for battery applications. We were surprised to find that the lithium can be removed reversibly down to at least  $x = 0.067$ . In the compositional range  $0.067 \leq x \leq 1$ , the layered structure appears to be kinetically stable at room temperature in the electrolyte.

#### Material Preparation and Characterization

A sample of nominal composition  $\text{LiCoO}_2$  was prepared by heating a pelletized mixture of lithium carbonate and cobalt carbonate in air at  $900^\circ\text{C}$  for 20 hrs followed by two further firings. An X-ray photograph ( $\text{CuK}\alpha$  radiation) taken on a Stoe focussing Guinier camera could be indexed on a hexagonal unit cell with  $a = 2.816(2)\text{\AA}$ ,  $c = 14.08(1)\text{\AA}$ . No extraneous reflections were observed, and the data agreed well with those of other workers (6).

Total cobalt was determined by EDTA titration and the cobalt oxidation state by reduction to  $\text{Co(II)}$  with excess ferrous ammonium sulphate and back titration with standard potassium dichromate. Analysis for oxidizing power gave 47.0%  $\text{Co(III)}$  by weight with total cobalt as 48.2%. This indicates a very small amount of  $\text{Co(II)}$  present, and the most satisfactory formulation is  $\text{Li}_{0.99}\text{Co}_{1.01}\text{O}_2$  with a small amount of cobalt atoms in the lithium layer.

#### Electrochemical Studies

The three-electrode assembly used for electrochemical characterization and low-temperature lithium extraction is shown in Fig. 2. The cathode consisted of about 20 mg of  $\text{LiCoO}_2$  powder pressed onto a 1-cm diameter stainless-steel gauze disc. The electrolyte was a 1 M  $\text{LiBF}_4$  solution in propylene carbonate that was soaked onto Whatman GF/D glass-fibre paper. The propylene carbonate had been dried by fractional distillation under vacuum and with molecular sieves. The cell was kept within an argon-filled KONTRON Vacuum Atmospheres glove box.

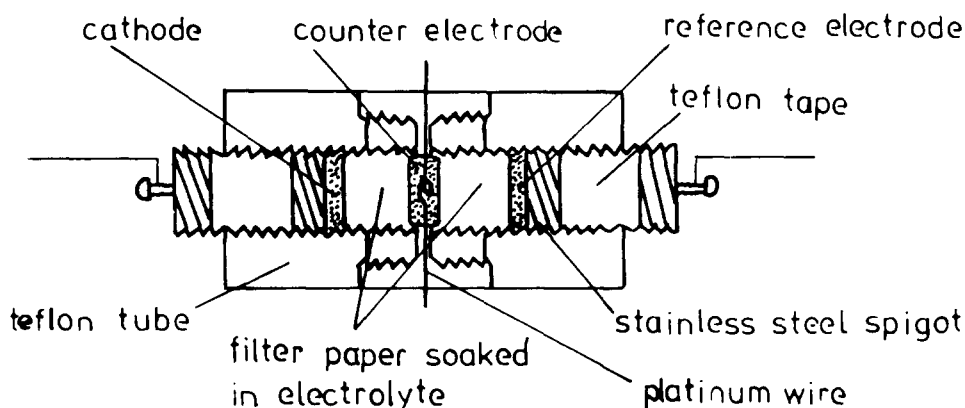


FIG. 2  
Cell for electrochemical measurements

Initially, lithium metal was used as both reference and counter electrodes, and a charging current was readily passed. As the measured open-cell voltages were found to be in excess of 4 V, there was a risk of electrolyte breakdown. To avoid this problem, both counter and reference electrodes of  $\text{Li}_{0.1}\text{V}_2\text{O}_5$  were used. These were prepared electrochemically in the cell from  $\text{V}_2\text{O}_5$ . They enabled measured voltages to be kept below 1.5 V (8). The compositions, as determined coulometrically at very low current density, indicated that lithium could be almost completely removed. The open circuit voltages shown in Fig. 3 were reproducible, and there was no indication of electrolyte decomposition. The E vs x curve corresponds to a theoretical energy density of  $1.11 \text{ kWh kg}^{-1}$  for the reaction  $0.923\text{Li} + \text{Li}_{0.067}\text{Co}_{1.01}\text{O}_2 = \text{Li}_{0.99}\text{Co}_{1.01}\text{O}_2$ .

Reversibility tests were carried out between  $\text{Li}_{0.99}\text{Co}_{1.01}\text{O}_2$  and  $\text{Li}_{0.5}\text{Co}_{1.01}\text{O}_2$  at current densities up to  $4 \text{ mA cm}^{-2}$  (Fig. 4). Overvoltages can be made low; they are sensitive to the cathode fabrication procedure. Optimum results were obtained by preparing a paste of the powder with propylene carbonate followed by drying and pressing onto a platinum plate. Current densities were somewhat lower below  $x = 0.5$  and also near  $x = 1.0$  where the electronic conductivity of the materials is lower.

The room-temperature chemical diffusion constant  $\bar{D}$ , estimated by the potentiostatic method of Weppner and Huggins (9), was  $\sim 5 \times 10^{-9} \text{ cm}^2 \text{ sec}^{-1}$  for x in the range 0.2 to 0.8. This is an approximate value as our electrode was not 100% dense and the active surface area was taken as the geometric area.

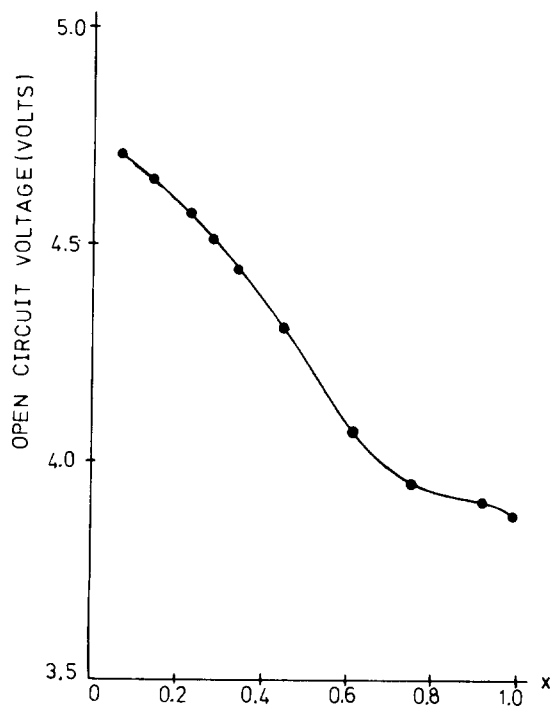


FIG. 3  
Open circuit voltage versus composition x  
for  $\text{Li}_x\text{Co}_{1.01}\text{O}_2/\text{Li}$

### Structural Studies

Small samples, corresponding coulometrically to  $x = 0.74$ ,  $0.49$  and  $0.33$ , were removed from the cell and examined by x-ray diffraction. Samples were pressed onto adhesive tape and stuck onto a glass slide in order to minimize any possible reaction with air or moisture. They were then transferred from the glove box to a Philips PW1050 powder diffractometer having a quartz analyzing monochromator. All peaks could be indexed on the hexagonal cell of  $\text{LiCoO}_2$ , and the intensity variation was that expected for this structure. Comparison of  $d_{\text{obs}}$  and  $d_{\text{calc}}$  and the refined lattice constants are given in Table 1. At  $x = 0.33$  peaks became fewer and broader; lattice parameters for  $x \leq 0.33$  are considered unreliable. Above  $x = 0.33$  there is a steady decrease in the a parameter and a small increase in the inter-layer c parameter as x decreases. The data imply a retention of the layer structure on removal of lithium atoms.

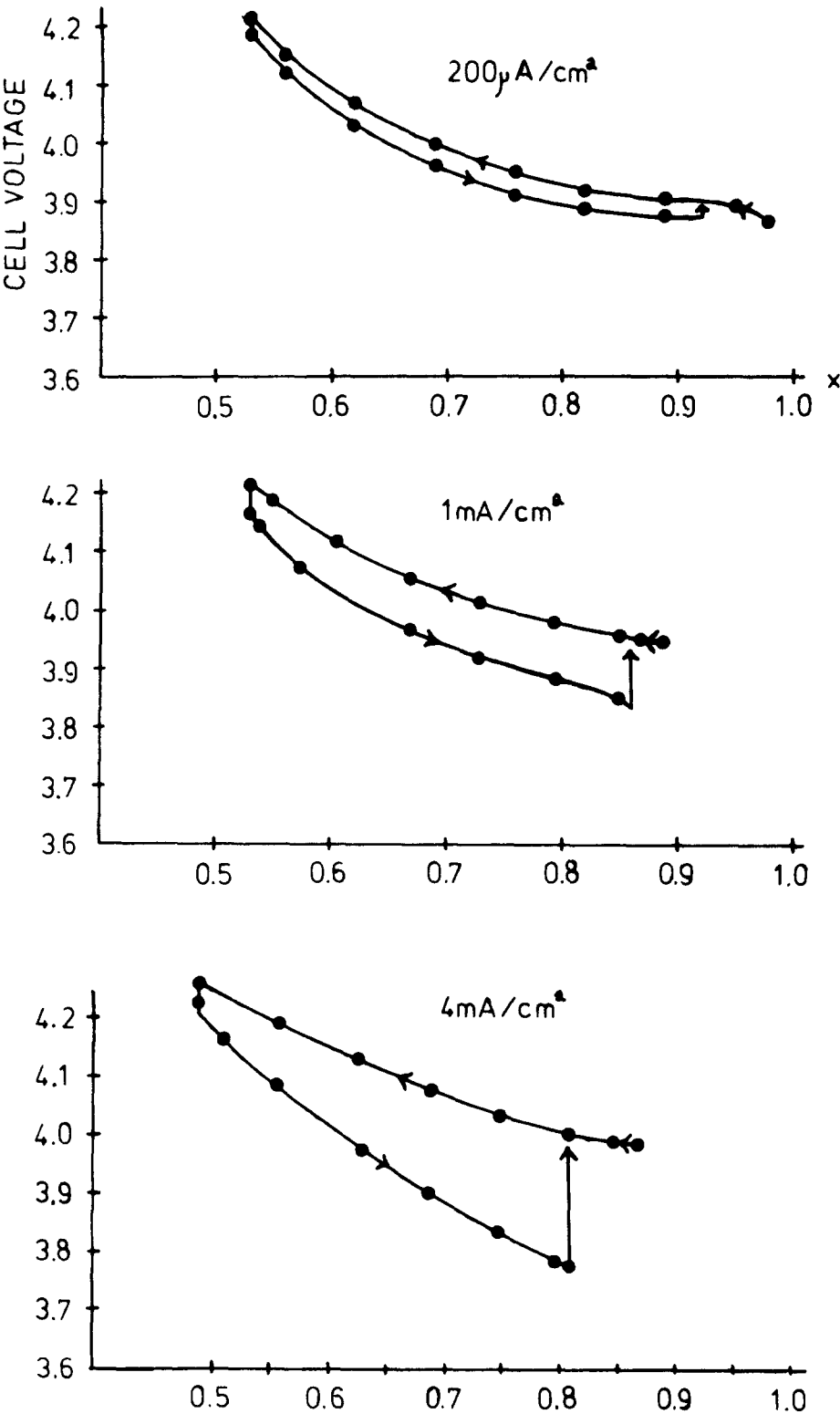


FIG. 4  
Voltage versus composition  $x$  for  $\text{Li}_x\text{Co}_{1.01}\text{O}_2/\text{Li}$  at various current densities

TABLE 1

X-ray Data for  $\text{Li}_x\text{Co}_{1.01}\text{O}_2$ 

| $\text{Li}_{0.74}\text{Co}_{1.01}\text{O}_2$ | dobs/ $\text{\AA}$ | dcalc/ $\text{\AA}$ | hkl |
|----------------------------------------------|--------------------|---------------------|-----|
|                                              | 4.720              | 4.719               | 003 |
|                                              | 2.398              | 2.397               | 101 |
|                                              | 2.368              | 2.365               | 006 |
| hexagonal                                    | 2.304              | 2.301               | 012 |
| a = 2.812(2)                                 | 2.006              | 2.007               | 104 |
| c = 14.22(1)                                 | 1.845              | 1.848               | 015 |
|                                              | 1.558              | 1.558               | 107 |
|                                              | 1.435              | 1.434               | 018 |
|                                              | 1.406              | 1.406               | 110 |
|                                              | 1.347              | 1.348               | 113 |
| $\text{Li}_{0.49}\text{Co}_{1.01}\text{O}_2$ | dobs/ $\text{\AA}$ | dcalc/ $\text{\AA}$ | hkl |
|                                              | 4.770              | 4.807               | 003 |
|                                              | 2.398              | 2.398               | 101 |
|                                              | 2.304              | 2.304               | 012 |
| hexagonal                                    | 2.014              | 2.016               | 104 |
| a = 2.807(2)                                 | 1.859              | 1.859               | 015 |
| c = 14.42(1)                                 | 1.603              | 1.602               | 009 |
|                                              | 1.573              | 1.571               | 107 |
|                                              | 1.447              | 1.448               | 018 |
|                                              | 1.404              | 1.404               | 110 |
|                                              | 1.347              | 1.347               | 113 |

Conclusions

1. The technique of topotactic electrochemical extraction of lithium from a known ternary transition-metal oxide has been demonstrated for the ordered-rocksalt structure of  $\text{LiCoO}_2$ . The fact that the  $\text{O}^{2-}$  ions form a close-packed array in this compound suggests that the method may well be applicable to a number of other systems.

2. Open-circuit voltages of 4-5 V for  $\text{Li}_x\text{CoO}_2/\text{Li}$  cells are consistent with the oxidizing power of the  $\text{Co(IV)}/\text{Co(III)}$  couple. The high voltages permit consideration of alternate negative electrodes, and two cell configurations become possible options.

- (a) Use of an insertion compound of low work function for the negative electrode and a liquid electrolyte. Such a cell would have a lower open-circuit voltage and a larger variation in voltage with state of charge.
- (b) Use of a solid electrolyte of large breakdown voltage to enable a greater fraction of the potential energy density to be utilized.

The practicality of these options has yet to be explored.

3. Overvoltages and reversibility appears promising for practical application, at least to  $x = 0.5$  and to current densities of  $1 \text{ mA cm}^{-2}$ . Three factors contribute to the higher  $\text{Li}^+$ -ion mobilities observed in  $\text{LiCoO}_2$  relative to preliminary data on other  $\text{LiMO}_2$  compounds:

- (a) The spacing between oxygen layers facing the Li layers is greater in  $\text{LiCoO}_2$  than in the other  $\text{LiMO}_2$  oxides for which sufficient x-ray data are available.
- (b) The high electron affinity of the low-spin  $\text{Co(IV)/Co(III)}$  couple makes the oxygen layers strongly polarizable toward the cobalt layers.
- (c) X-ray powder refinements (10) indicate almost complete ordering of lithium and cobalt into alternate (111) layers; this is not true of the corresponding nickel compounds.

4. The practical characteristics of  $\text{Li}_x\text{CoO}_2$  as a cathode material also depend on extrinsic properties, such as particle size and electrode porosity, and hence on the method of preparation.

Further characteristics of the intrinsic and extrinsic properties of this new system are being made.

#### Acknowledgments

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